

Evaluating Machine Learning Models for Cyber Incident Prediction in UK Data Breach Reports

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1.0 Abstract

This paper explores the use of machine learning to predict self-reported data breaches using a publicly available dataset from the UK Information Commissioner's Office to serve as a guidance for legal professionals. Several classification models were tested and evaluated, including Logistic Regression, Extreme Gradient Boosting (XGBoost), Random Forest and Multi-Layer Perceptron (MLP). XGBoost achieved the highest F1 score of 0.645, indicating a moderate performance with room for improvement. The model also achieved an accuracy score of 87.1%, demonstrating its effectiveness in classifying whether a breach was considered a cyber incident. The evaluation metrics suggested that the model is good at finding non-cyber incidents and struggles with finding cyber incidents likely due to limitations in the data. These findings display that there is potential for machine learning support in regulatory bodies if there is strong data to support it.

2.0 Research Objectives

Q1: Can a machine learning model accurately predict whether a self-reported data breach will be classified as a cyber incident?

Q2: Which sectors are most likely to be a cyber incident?

Q3: Which machine learning model handles this kind of scenario best?

Q4: Is machine learning a viable option in a law setting when there are data restrictions due to sensitivity.

Hypothesis: Logistic Regression will perform best due to the nature of the ICO dataset due to the tabular structure of the data and the algorithm will be a classification model. As the research involves legal data, logistic regression provides model interpretability through coefficient analysis.

3.0 Introduction

In the United Kingdom, the Information Commissioner's Office (ICO) handles data protection regulations and data mishandling by organisations. The ICO publishes a publicly available data set each quarter of the year for data breach decisions which is provided under the Open Government Licence v3.0 (Information Commissioner's Office, 2025) . The Open Government Licence allows anyone to openly adapt and exploit the information, giving grounds for this paper to analyse and use the data within an academic practice (Open Government License, no date). The goal of this project is to explore how predictive modelling can enhance regulatory bodies decision making in a lawful environment. By using machine learning techniques, we aim to identify patterns and uncover relationships between specific sectors and their likelihood of being victim to a breach.

4.0 Literature Review

4.1 Literature Selection Criteria

Academic papers were sourced using Google Scholar and IEEE Xplore. Relevant search terms included ‘Machine Learning in a law environment’, ‘Explainable AI’, ‘Ethics of AI in law’ and ‘GDPR and Artificial Intelligence’. Preference was given to papers published after 2018 to ensure the contents aligned with the GDPR regulations. The selection of these papers was based on citation count and relevance to the proposed research topic.

4.2 Machine Learning in Legal Sectors

As technology advances, machine learning has become a staple instrument in conducting tasks in many sectors, including law and data protection. With the introduction of the General Data Protection Regulation (GDPR) and the Data Protection Act in 2018, handling sensitive data has become a crucial responsibility for organisations (GOV UK, 2018). The GDPR law enforces strict regulations on how personal data is collected and handled to ensure public security and responsibility is bestowed upon the organisations that handle the data. When considering the inclusion of machine learning in an area such as law, it is critical to consider data protection legislation as much of the data being used can be classified as sensitive.

A paper published in March of 2023 by Lore et al. explores the idea of artificial intelligence in GDPR decisions (Lore et al, 2023). The study proposes an Artificial Intelligence framework that uses language processing techniques and machine learning algorithms to help assess whether public Italian documents align with the GDPR laws. This paper was chosen due to its research involving a classification model applied within legal context, similar to the focus of this study. The authors designed an AI Framework called “INTREPID”, which was created to detect potential GDPR breaches. Lore points out that the current method that the Italian Public Administration have adopted to analyse documents is done manually, opening the door for potential improvements and optimization through AI and machine learning automation.

Using the model to read documents to identify whether they aligned with GDPR regulations, the model had an F1-score of 0.79 using a logistic regression algorithm. While Lore et al. achieved an F1 score of 0.79, this papers research shows XGBoost (F1=0.645) outperformed Logistic Regression (F1=0.518) with the ICO dataset, suggesting that the complex non-linear relationships between sectors and breach types are better captured by ensemble methods.

This research provides a strong foundation for the concept of machine learning and artificial intelligence utilization in a law environment.

4.3 Ethics of Machine Learning in Legal Sectors

In contrast to the research conducted by Lore et al, which served as a binary GDPR compliance model, consideration to the limits of machine learning in legal scenarios that involve real life consequences questions the ethics of such practice. In a paper that explores this, the author Clare Barale argues that although predictive models may help legal professionals in identifying patterns and trends in data, they risk oversimplifying the complex nature of human decision making. Barale points out that justice is not solely data-driven and warns against total reliance on algorithms in spaces where human empathy and fairness are required (Barale, Rovatsos and Bhuta, 2025). Navigating the law can be a complex and situational endeavour that is inherently dependent on human decision making despite the potential of machine learning. Ethical concerns raise a valid issue in the prospect of machine learning in law while not discrediting its usage for efficiency and should be considered when undertaking potential life changing decisions.

Using the research conducted within both papers, we can build a strong argument for using machine learning to optimize tedious tasks like the GDPR compliance algorithm while still upholding ethical integrity by limiting the scope of the predictive model. By introducing the INTREPID framework and considering the ethics of such practice, the authors support the idea of AI optimization like the ICO model under the premise that it serves only as a tool for guidance and not a final judgement.

4.4 Explainable Artificial Intelligence (xAI)

Artificial Intelligence has been well established in the technological world. As advancements are made in the capabilities of AI, more industries are adopting its use in some way or another. Despite the advantages that AI and machine learning can bring, using it in an environment where there are real world consequences is a trivial task. While AI models like INTREPID display impressive results, it is crucial to consider the explainability of them, especially in fields like law.

In the book, Natural Language Processing and Chinese Computing, the author makes some interesting points about the explainability of modern AI models

(Xu *et al.*, 2019). If a person can interpret the input and output of a model, but there is a lack of explanation on how the input became the output, then using these techniques in consequential circumstances becomes complicated. This is applicable to both the proposed ICO classification algorithm and the INTREPID model. The INTREPID model was built using a logistic regression algorithm. Logistic Regression is a good model when considering explainable AI which is why it will be one of the models selected for the ICO data (Dwivedi *et al.*, 2023).

While Logistic Regression is a good option due to its understandable mechanics, it remains important to look for methods to explain when using any kind of machine learning algorithm.

4.4.1 SHAP

SHAP (Shapely Additive exPlanations) is a XAI process that is based on a game theory. SHAP works by distributing a feature value that represents the contribution it has to the model's prediction (Lundberg and Lee, 2017). The SHAP method calculates the impact that each feature has on the final prediction by considering all possible combinations of features and averaging the marginal contribution of them across the combinations. In predictive modelling analysis, SHAP has been shown to outperform LIME in several metrics. According to Velmurugan, SHAP offers near perfect stability and is computationally more efficient (Velmurugan *et al.*, 2020). As XGBoost, this project's chosen model, is a complex ensemble of decision trees, SHAP accounts for all possible combinations of features ensuring an accurate and consistent allocation of each features contribution to the prediction.

As the ICO research is conducted using real world cyber breach data, it is vital to consider AI Explainability.

5.0 Methodology

5.1 Introduction

The goal of the ICO machine learning algorithm is to analyse the data and predict the outcome of self-reported data breaches. Using the visualizations, we can find trends in the data and ultimately predict whether the breach was a cyber incident or not. Using the patterns and commonality found between each sector in relation to the cyber incident column, data backed predictions can be made.

This section details the methods that were used to create the proposed machine learning algorithm.

5.2 Dataset

ICO

Data:

<https://ico.org.uk/action-weve-taken/complaints-and-concerns-data-sets/self-reported-personal-data-breach-cases/>

A screenshot of the data used in this project is provided in **Appendix A** to illustrate the data structure and features.

To ensure that the machine learning algorithm performs adequately, it is important to provide a quality dataset. Choosing good data for a machine learning algorithm is crucial, as poor data quality can lead to false results and low performance (Sharma, 2024). A paper written in 2021 reinforces this idea, referring to the saying, “garbage in, garbage out” (Geiger *et al.*, 2021). The authors use this term to highlight the fact that a machine learning model is only as good as the data you give it, meaning that despite the potential of the algorithm, if the data is arbitrary then the results will be similar.

Before developing the machine learning model, a data set was sourced from the official ICO website. The data details self-reported data breaches submitted to the United Kingdom’s Information Commissioner’s Office. The dataset totals 3,416 cases and is taken directly from the ICO website, meaning there is enough data to train the model and uphold integrity due to the trustworthy source.

Although the data set provided an ample amount of information for usage in a machine learning algorithm, there were several things that hindered the potential of the model. The target column, ‘ISCyberIncident’, which is binary with each value either being “Yes” or “No” had a large imbalance with 80% of the values being “No”. Along with this, most other features did not contain valuable data

that a machine learning algorithm would typically use. To combat this, new features were created using the “Sector” and “Sub Sector” values. Two features called “Sector Percentage” and “Sub Sector Percentage” were engineered by calculating the percentage of cases that were cyber incidents for each sector and sub sector.

5.2.1 Preparing the Data

Before going further in development of the machine learning model, data preparation is essential to ensure sufficient training and accurate results.

Referring to a paper written by Ndung’u Rachael Njeri from Murang’s University of Technology, there are four key steps for data pre-processing in preparation for a machine learning algorithm (Njeri, 2022).

1. **Data Collection** – Gathering high quality data from a trustworthy source.
2. **Data Cleaning** – Dealing with missing values and removing or correcting errors.
3. **Data Transformation** – Splitting and normalization of data and converting data into a suitable format for the model.
4. **Data Reduction** – Removing arbitrary data that has no meaningful contribution to the model’s development and results.

Using these four steps with the ICO data will ensure that it is ready for training and testing for the model. As the ICO data is from a reputable source, the data collection aspect has been fulfilled. Despite there being multiple data sets available on the ICO’s website, each containing data breaches for every quarter of the year, the single data set will be enough for the scope of this project, meaning we do not have to perform any merging of data.

Data Cleaning

To uncover any missing values, the sum function was used to count the null values in the dataset. The sum returned zero for each column, meaning no missing values were present.

Data Transformation

Before training the data through the model, it is important to transform it into a format that the algorithm can process. The data was first split into 80 training and 20% testing to evaluate the models’ performance on unseen data. The

standard scaler from the sci-kit learn library was the used on the training and testing sets to normalize the features consistently.

Data Reduction

The data set is large and comes in 19 different columns. When creating a machine learning model, it is important to consider efficiency as well as other markers of success like accuracy. A common way to enhance efficiency is by removing columns that are arbitrary to the prediction.

Dropped Columns:

- **ICOCASERef** – an ID/Reference number – adds no value.
- **All Date columns** – Dates not needed as they all take place around similar times.

5.2.2 Data Analysis

Once the data has been pre-processed and is ready for the model, it is good practice to have a closer look before the training process. This allows us to see if there are any patterns or insightful information we can derive from it, giving more clarity to the model's results when it comes to making predictions.



Figure 1. Sector Cyber Incidents percentage chart

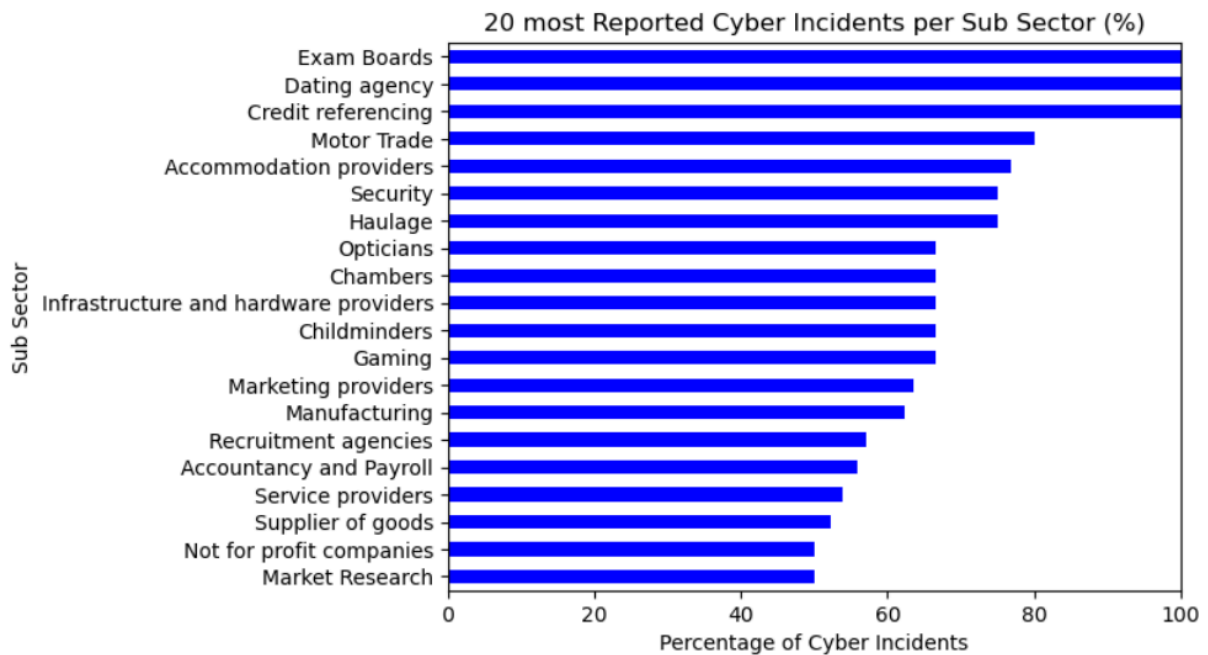


Figure 2. Sub Sector Cyber Incidents percentage chart

The above charts display the percentage of reported breaches that were found to be cyber incidents per sector and sub sector. This was accomplished by dividing the total number of cyber incidents by the total self-reported breaches for each sector and multiplying by one hundred to return a percentage.

The marketing sector emerges as the most frequently affected with retail and manufacture being closely behind. This suggests that organisations within these sectors face heightened risk of cyber incidents, potentially due to their reliance on consumers data and communication. The same methodology can be applied to the subsectors, with this data giving a deeper insight into the risk analysis due to the higher percentages and more specific statistics.

This information is valuable, as the ICO prediction model is designed to serve only as a guide for legal professionals to enhance judgement, knowing which sectors are most vulnerable can validate decision making if the sector or subsector is statistically prone to attack.

5.3 Model Selection

Using a Stratified K-Fold cross validation with 5 splits, Extreme Gradient Boosting, Logistic Regression, Random Forest and a Neural Network were all trained on the data to see which one performed best. A stratified K-fold cross validation was used to avoid bias from a single random train/test split and retaining original class distribution in each fold, helping address the imbalanced data set. This ensured all the data was used for training and testing, making the results more reliable.

Model	Mean F1 Score	Std. Deviation
XGBoost	0.645	(+/-) 0.0381
Logistic Regression	0.518	(+/-) 0.0311
Random Forest	0.614	(+/-) 0.0300
Neural Network	0.5182	(+/-) 0.0176

Extreme Gradient Boosting has the highest F1 score with 0.645, outperforming the other three models. This suggests that XGBoost is the most effective at classifying cyber breaches in an imbalanced dataset. In addition to the F1 score, the standard deviation ± 0.0381 suggests that the model is reasonably consistent across all data splits, however there is room for improvement. In comparison to the XGBoost model, the Logistic Regression algorithm seems to have underperformed which contrasts the hypothesis of this paper. This may be due to the lack of linearity in the dataset, with limited clear relationships and the presence of complex patterns.

6.0 Evaluation

This section evaluates the performance of each machine learning models used which were assess using F1-Score, accuracy, precision and recall to suitability and predictive performance in the context of the dataset.

6.1 XGBoost Evaluation

Model Performance Metrics

Metric	Value	Explanation
Accuracy	0.8707	Identified both “Yes” and “No” outcomes 87.07% of the time.
Precision (Yes)	0.7525	Of all predicted “Yes” cases, 75.25% were true positives.
Recall (Yes)	0.5507	Of all actual “Yes” cases, 55.07% were correctly identified.
F1 Score (Yes)	0.6360	Overall F1 score on the testing data.

The model’s performance metrics indicate a strong accuracy score of 87.1%, suggesting that it is generally reliable in predicting outcomes.

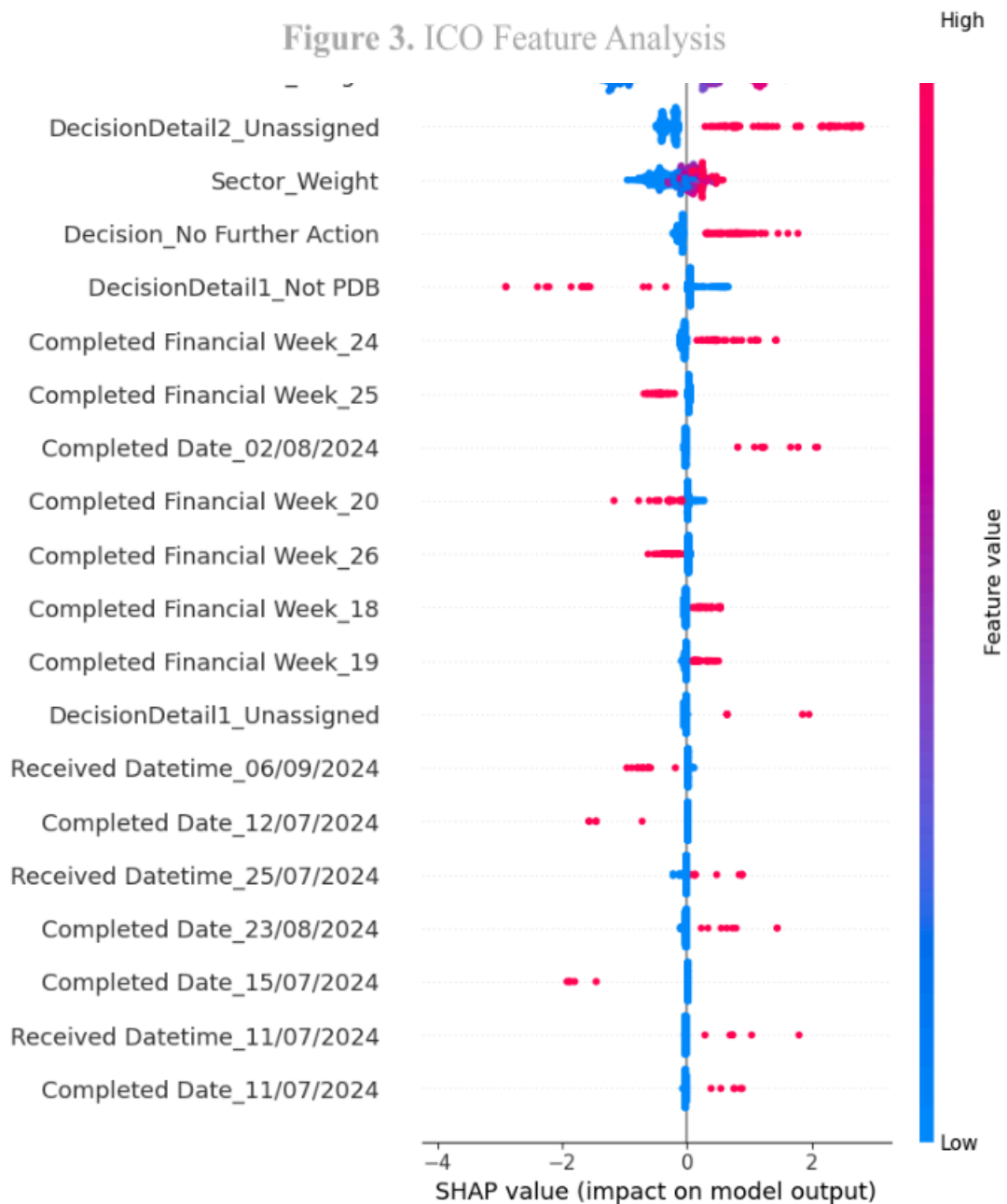
The precision score Of 0.75 also suggests that when the model predicts a cyber incident, it is correct 75% of the time.

Despite these high metrics, the recall score of 0.5507 shows that the model only identifies 55% of actual cyber incidents, meaning many are missing.

The F-1 score which balances the recall and precision highlights the moderate performance of the model and suggests that there is room for improvement.

These results display the models’ high capabilities of identifying non-cyber incidents rather than cyber incidents, which can be explained by the imbalance in the data set.

6.2 SHAP Analysis



As emphasised in the literature review of this paper, being able to explain the inner workings of a machine learning model is vital. A SHAP analysis was chosen to visualize the importance of each feature towards the final predictions, giving insight into how the model makes decisions.

The analysis shows that the SubSector_Weight, DecisionDetail2_Unassigned and Sector_Weight were the top three features driving predictions. With the sector weight being an important feature in the predictions, sub sector weight goes even deeper, alluding to the fact that although a sector like “Healthcare” and its weight value contribute to the prediction, the sub sectors within healthcare and their weights contribute more. This is due to the specificity of subsectors that capture more nuance patterns and can help differentiate between outcomes.

The DecisionDetail2_unassigned feature is flagged by the model due to the ambiguity of the data, serving as a critical signal for cases requiring human intervention. If the model encounters an unassigned label, it interprets the undefined classification as a risk indicator.

7.0 Conclusion

The classification model demonstrates good overall performance with an accuracy of 87.1%. This suggests general effectiveness in predicting outcomes to data breaches. Despite the high accuracy score, the F1 Score highlights the underwhelming reality, indicating that nearly half of the actual cyber incidents were missed as identified by the recall score. The imbalance between the recall and precision indicates that the model is cautious about labelling a case as a cyber incident in the risk of it being a false positive, resulting in missing genuine cases.

This study addressed four primary objectives. First, it found that machine learning can moderately classify self-reported data breaches, achieving a high accuracy with limitations in recall. Secondly, the analysis found that Marketing, Retail and Manufacturing and Online Technology and Telecoms were the most common sectors to be victim of a cyber incident. Third, comparison across the chosen algorithms found that XGBoost performed best which contrasts the hypothesis of this project where Logistic Regression was expected to be the most optimal. Finally, while the research determined that machine learning can assist legal professionals with decision-making and pattern recognition, practical application is still limited by data limitations.

Overall, the model serves as a good base for further research to be built upon. To use such methods in a real-world scenario, the model would require a larger and more balanced data set with more practical features. The findings contribute to the idea of machine learning in the legal sector, while emphasizing the flaws of the concept.

8.0 Resources

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9.0 Appendix

Appendix A: Screenshot of raw ICO data

Case Statu	Legislation	ICOCASE#	Received	Received	Received	Received	Received	Completed	Completed	Completed	Completed	Completed	Completed	Completed	Sector	SubSector	Submitted	Decision	DecisionD	DecisionD	ISOCyberIncident
Completec	GDPR	IC-316312	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Health	General Pr	Anon DC	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316329	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Health	Healthcar	Anon DC	Informal at	Breach rec	Advice give	Yes	
Completec	GDPR	IC-316333	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Charitable	Local char	Battersea	No Further	Breach rec	Unassigne	Yes	
Completec	GDPR	IC-315694	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Membersh	Clubs	Bilston Swi	Informal at	Breach rec	Advice give	Yes	
Completec	GDPR	IC-316607	#####	2024	2024/25	Q2	04-Jul	#####	2024	2024/25	Q2	04-Jul	14	Membersh	Clubs	British Eve	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316534	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Education	Maintainet	Carr Hill H	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316295	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Finance, ir	Financial s	Caxton Pay	No Further	Breach rec	Unassigne	Yes	
Completec	GDPR	IC-310155	#####	2024	2024/25	Q1	02-May	#####	2024	2024/25	Q2	04-Jul	14	Finance, ir	Insolvency	CG Recove	No Further	Breach rec	Unassigne	Yes	
Completec	GDPR	IC-316591	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Health	Secondary	Chiltern Su	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316454	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Retail and	Supplier of	Credenhill	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316339	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Social care	Domicilian	Deafway	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-301202	#####	2024	2024/25	Q1	01-Apr	#####	2024	2024/25	Q2	04-Jul	14	Local gove	Unitary Aut	Derby City	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316359	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Health	Pharmacis	DialAChen	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316584	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Transport	Accommo	Ennismore	Informal at	Breach rec	Advice give	Yes	
Completec	GDPR	IC-316446	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	General bl	Unassigne	Fulham Fo	Informal at	Breach rec	Advice give	Yes	
Completec	GDPR	IC-315379	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Education	Academy	Gordon's S	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316360	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Finance, ir	Financial s	Isio Group	Informal at	Breach rec	Advice give	Yes	
Completec	GDPR	IC-316367	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Finance, ir	Financial s	Isio Service	Informal at	Breach rec	Advice give	Yes	
Completec	GDPR	IC-316379	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Finance, ir	Financial s	Isio Total R	Informal at	Breach rec	Advice give	Yes	
Completec	GDPR	IC-316376	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Finance, ir	Financial s	Isio Wealth	Informal at	Breach rec	Advice give	Yes	
Completec	GDPR	IC-316084	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Charitable	Local char	Kingston V	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316375	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Legal	Legal prof	Lamb Broo	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316392	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Local gove	Unitary Aut	London Bo	Informal at	Breach rec	Advice give	No	
Completec	GDPR	IC-316457	#####	2024	2024/25	Q1	03-Jun	#####	2024	2024/25	Q2	04-Jul	14	Local gove	Unitary Aut	London Bo	Informal at	Breach rec	Advice give	No	